

Jaume **Ivars Grimalt**

Machine Learning Research | Computer Vision | Medical Imaging

Currently in Valencia, Spain | EU citizen | Open to relocation to Zurich, Munich, London

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Profile

Computer vision engineer moving into research. My current work studies representational dynamics in residual vision networks: how features evolve with depth and what that reveals about weight sharing and model internals. I have applied deep learning to medical and biological data, from brain MRI segmentation at MIALAB (UPV) to a microbiology detection system I built from scratch and tested in five laboratories.

Education

M.Sc. Artificial Intelligence, Pattern Recognition & Digital Imaging — *Valencia, Spain*

UPV · GPA 8.7/10 · Thesis Honours (became Mycrospace) — *2023 – 2024*

Technology Management — *Valencia / Munich*

CDTM (TUM/LMU) · 16 selected — *2024*

B.Sc. Computer Engineering — *Valencia, Spain*

UPV · GPA 8.3/10 · Honours in ML and Statistics — *2019 – 2023*

Research

• *What Extra Euler Steps Reveal About Residual Networks: Feature Dynamics in Weight-Tied and Untied Stages.* Manuscript in preparation. jaumeivars.com/research/representation-dynamics. Studies continuous-depth dynamics in ConvNeXt, Swin and ResNet: finer numerical integration without retraining, trajectory geometry and per-block weight corrections. With **~9.6M** fewer parameters, a weight-shared ConvNeXt-T stage trained from scratch on ImageNet-1k reaches **80.34%** top-1 (baseline: **80.96%**).

Publications

• *Auto-N2N: Self-supervised MRI Denoising using Synthetic Pair Generation and Mixture of Experts.* S. Morell-Ortega et al., incl. **J. Ivars Grimalt**. Preprint, 2026. OpenReview

• *Detecting Cytopathic Effect in Cell Culture with Computer Vision.* W. Ragazzo, **J. Ivars Grimalt**. Under review.

Experience

Mycrospace — *Valencia, Spain*

Co-Founder & CTO — *2024 – Oct 2026*

- Started as a Master's thesis in 2024. The company was founded in January 2026. Computer vision for microbiology, tested in **5 labs**. 1st Prize IdeasUPV (2024).
- **Model.** Own ConvNeXt detector (YOLO/DQ-DETR-inspired), from scratch; 10–2000 colonies/image. **67 mAP50-95**; 92% recall / 93% precision.
- **Training. 12,500+** GPU-hours on Leonardo (two EuroHPC AI Factory allocations). **4x A100**, DDP/AMP/torch.compile/SLURM, checkpoint-resumed jobs. Own framework: k-fold CV, WBF ensembles.
- **Data. 1,500/5,000** images annotated with microbiologists. Weakly-supervised labelling (click → crop → 4-ch CNN heatmap → bbox); versioning + k-fold label scoring.
- **Product/infra.** BullMQ/Redis inference on GCP, plus NestJS/PostgreSQL, Next.js and React Native.

MIALAB: Medical Imaging Analysis Laboratory, UPV — *Valencia, Spain*

Research Engineer (full-time) — *May 2025 – Jan 2026*

- Volumetric brain MRI segmentation with nnU-Net; multi-scale feature strategies. Profiled volBrain (**700k+** volumes) from NIfTI ingest to voxel prediction for inference cost.

Earlier experience: Tapstar, Full-Stack Engineer (freelance), 2025; Vesta-Z, Data Engineer, 2023–2025; Neurocatching, Data Scientist Intern, 2023; Sciling, Backend Developer Intern, 2022.

Technical Skills

AI & ML	Object detection, instance segmentation, density estimation, classification, CNNs, ViTs, foundation models, Neural ODEs, PINNs
Medical imaging	nnU-Net, volumetric MRI, NIfTI
Training	Python, PyTorch (DDP, AMP, torch.compile), Hugging Face, timm, multi-GPU HPC (SLURM)
Production	ONNX, Docker, CI/CD

Languages

Spanish (native), Catalan (native), English (C1)